

Seong-Jun Kang, Ph.D.

Computational immunologist working in single-cell and spatial multi-omics of the immune system

Research Scientist applicant · Hyman Lab, MassGeneral Institute for Neurodegenerative Disease · transcriptomics and neurodegeneration

101 Canal Street
Boston, MA 02114
(551) 239-0341
sjkang89@gmail.com

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U.S. work authorized · no sponsorship required

PROFESSIONAL STATUS

I'm an immunologist working in computational single-cell analysis. I read immune cell state from large, incomplete single-cell and spatial data, and I check that it holds on held-out data. The brain work I have done is single-cell immune profiling in an Alzheimer's mouse model, where I built the path from tissue harvest through 10x to the analysis.

RESEARCH OBJECTIVE

to add an immune-cell layer to the transcriptomic dissection of the human Alzheimer's brain, and to separate immune programs that drive neurodegeneration from those that only track it. I write my own funding: lead proposal writer on ~\$1.3M of funded industry grants and a >30% contributor to the ~\$1.6M SCAID atlas proposal.

SCIENTIFIC INTERESTS

single-cell and spatial multi-omics; immune cell-state inference from incomplete data; consensus NMF and isoform-resolved analysis; neuro-immune interactions in neurodegeneration; atlas-scale QC across many donors and sites; calibrated, held-out-validated ML.

■ WHAT I WOULD BRING

- **Brain immune single-cell analysis.** In the 5x FAD Alzheimer's mouse model I built the analysis path from end to end, running whole-brain harvest, enzymatic digestion, Percoll enrichment, and FACS into 10x and then the analysis. It resolved ten immune populations from 11,587 cells and showed exhausted CD8 tissue-resident memory T cells engaging the choroid plexus epithelium at the blood-CSF barrier, while that epithelium turned on MHC class I, interferon-stimulated genes, and JAK-STAT signaling (Journal of Neuroimmunology 2025, first author).
- **Cell-state calling and held-out testing.** From 251,055 memory CD4 T cells with paired T-cell receptors I used consensus NMF to find twelve reproducible gene programs. Five-prime single-cell reads cannot separate the short and long PDE4D isoforms, so I wrote a UMI-aware pipeline that goes back to the aligned reads and assigns the isoform per cell. The short-isoform state recurs across 702 samples and 23 conditions in the SCAID atlas, and a classifier on subset composition reaches a held-out AUC of 0.78. The integration, the NMF, the isoform pipeline, and the classifier are my work, and colleagues carried out the wet validation (co-first author, in revision at Nature Immunology).
- **Methods that transfer.** Olink targeted plasma proteomics, GeoMx region-of-interest microdissection and profiling, and high-parameter flow panels are assays I have run, and I designed the causal overexpression test in the PDE4D study. I ran them on immune cells and tumor tissue, and I have not worked with human brain tissue. What transfers is the method rather than the biology, and the biology is what I would be learning from the people who already do it.
- **A first six months.** In the first three months I would take one human brain single-nucleus dataset the group already has and annotate the immune and perivascular compartment, writing the marker logic down and reporting uncertainty per call, with no new wet work. In the next three months I would test the calls on held-out donors, run consensus NMF for the programs, and check whether program scores track pathology stage rather than tissue batch or postmortem interval. The go/no-go at six months is whether the calls recover states already known from sorted-cell references and whether they hold out of sample.
- **Grants written.** In industry I was the lead proposal writer on two Korea Drug Development Fund awards for an anti-CD40 antibody (about \$571,000 and \$321,000) and on two Ministry of SMEs and Startups awards, and I did the proposal and study design for a NAVER-funded engineered Treg program. Those come to about \$1.3 million of funded work. During my doctorate I wrote more than 30 percent of the proposal for SCAID, the Korean single-cell atlas of immune disease, funded at about \$1.6 million, and I then ran its multi-site QC, milestone tracking and multi-PI coordination. All of these were held by the institutional principal investigator, so my role was proposal writing and project execution, not as PI.

EDUCATION

DEGREES

Ph.D. 2014–2023	Seoul National University , Seoul, Republic of Korea Department of Biomedical Sciences <i>Thesis: Exploration of the Role of Regulatory T cells in Various Pathogenic Contexts</i> Advisor: Chung-Gyu Park, M.D., Ph.D.
M.S. 2012–2014	Seoul National University , Seoul, Republic of Korea Department of Biomedical Sciences <i>Thesis: Enhanced LPS-Induced Monocyte Activation by Recombinant Human Soluble CD14</i> Advisor: Chung-Gyu Park, M.D., Ph.D.
B.S. 2008–2012	Sungkyunkwan University , Suwon, Republic of Korea Department of Genetic Engineering

RESEARCH & PROFESSIONAL EXPERIENCE

Researcher (postdoctoral) 2025–present	Yonsei University Wonju College of Medicine · Dept. of Physiology / Organelle Medicine Research Center, Wonju, Republic of Korea Deep-learning multimodal-integration models fusing imaging, transcriptomic, and clinical data into single predictive models.
Associate Director 2023–2025	PB Immune Therapeutics Inc. · Dept. of Basic Research, Seoul, Republic of Korea Led a humanized anti-CD40 antibody program: candidate ranking by surface plasmon resonance (Biacore) and ELISA, receptor-blocking confirmation, human and non-human-primate cross-reactivity, in vivo efficacy in primates, and the regulatory documentation. Placed and oversaw CRO and core-facility work, from study design and assay conditions through data delivery and go/no-go calls. Built the in-house single-cell and spatial analysis pipeline. Ran cynomolgus and marmoset EAE studies for multiple sclerosis programs.
Manager 2021–2023	PB Immune Therapeutics Inc. · Dept. of Basic Research, Seoul, Republic of Korea Wrote the two KDDF proposals that funded the anti-CD40 program and ran the lead-selection work under them.
Research Student 2020–2023	SCAID, Single Cell Atlas of Immune Diseases , Genomic Medicine Institute, Seoul National University, Seoul, Republic of Korea Contributed the founding proposal; executed within the funded program (700+ donors × 8 organs): milestone tracking, multi-site QC, multi-PI coordination.
Researcher 2019–2021	Xenotransplantation Research Center, Seoul National University , Seoul, Republic of Korea Rhesus macaque tissue processing and longitudinal immune monitoring across transplantation studies.
Teaching Assistant 2014–2016	Seoul National University College of Medicine · Dept. of Microbiology and Immunology, Seoul, Republic of Korea

TECHNICAL EXPERTISE

Brain and CNS tissue	Mouse brain immune profiling run end to end: whole-brain harvest, enzymatic digestion, Percoll enrichment, CD45 ⁺ FACS, 10× library generation, and the analysis (<i>J Neuroimmunol</i> 2025, first author; 5×FAD model, 11,587 cells, 10 immune populations). Neuro-immune analysis of choroid plexus epithelium and infiltrating CD8 ⁺ resident-memory T cells at the blood-CSF barrier. Non-human-primate CNS tissue, brain and spinal cord, across rhesus, cynomolgus and marmoset studies, including EAE models of multiple sclerosis.
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Cell-state calling and its checks	Consensus NMF gene programs ($k = 14 \rightarrow 12$ reproducible programs across 251,055 memory CD4 ⁺ T cells). Custom UMI-aware, BAM-level isoform code (Python) resolving PDE4D short- versus long-isoform identity from 5' reads, which a count matrix cannot separate. Milo KNN-graph differential abundance; scVI/scANVI integration; condition classifiers reported with held-out AUC and confidence intervals. Multi-site sample QC, milestone tracking and multi-PI coordination across a 700-donor, 8-organ atlas (SCAID).
Binding & protein interaction	Surface plasmon resonance (Biacore) run hands-on for antibody lead work: association and dissociation rates, affinity ranking across a candidate panel, receptor-blocking confirmation, and human / non-human-primate cross-reactivity. ELISA -based affinity and epitope analysis; competition binding. Antibody lead selection and optimization from candidate panel to registered patent. Functional validation of a peptide inhibitor of calcineurin in primary human T cells (CABIN1 core peptide, <i>Exp Mol Med</i> 2022).
Program & external execution	CRO and core-facility work from study design and placement through data delivery and go/no-go calls; assay conditions specified to the vendor; regulatory documentation for a therapeutic antibody program; funder milestones and reporting; multi-site sample QC across a 700-donor collection.
Cell, tissue & immunological assays	Primary human T-cell isolation, expansion and functional assays; monocyte and dendritic-cell differentiation; mesenchymal stem cell culture. Enzymatic tissue digestion, Percoll enrichment, cell and nucleus isolation. High-dimensional flow cytometry (panels up to 24 markers); multiplex panel design; 10-color 6-way FACS. Olink targeted plasma proteomics (immune & neurology panels).
Single-cell / spatial / multi-omic	scRNA-seq, scTCR-seq, snRNA/snATAC; 10x 5' / 3' library generation and 10x Chromium Connect automated preparation. Seurat , Scanpy , Squidpy , Monocle3 , CellChat/NicheNet ; scVI/scANVI, batch integration, consensus NMF ; Milo KNN-graph differential abundance, GLIPH, InferCNV. Custom UMI-aware, BAM-level isoform analysis code (Python). Spatial: GeoMx DSP run end to end, Visium , MACSima cyclic multiplexed immunofluorescence.
Computational	Python: scikit-learn, XGBoost/LightGBM/CatBoost, PyTorch ; AUC-validated classifiers held out on independent cohorts; multimodal deep-learning integration (imaging + transcriptomic + clinical). Linux servers · R/RStudio · Git-based reproducible analysis.
Vision-language and retrieval models	Finetuning . LoRA adaptation of an open vision-language model (Gemma 4 12B , mlx-vlm) so that an amyloid pathology tile returns a structured JSON reading, with the schema enforced during decoding and the split held out at the case level. Against the same model zero-shot on 800 held-out tiles, cored-plaque F1 went from 0.12 to 0.44 and cerebral amyloid angiopathy from 0.00 to 0.69. Retrieval . A retrieval-augmented question-answering service written end to end: corpus chunking, BM25 and dense-vector retrieval fused by reciprocal rank, query expansion, and streamed answers that carry their sources. FastAPI , containerized, on Google Cloud Run ; it answers questions about this CV at kangseongjun.com.
Microscopy & animal models	Spinning-disk confocal (Nikon CSU-W1 SORA), point-scanning confocal (Leica SP8), two-photon; IHC/IF (frozen & FFPE) with quantitative image analysis. Non-human primates across three species (rhesus, cynomolgus, marmoset), including EAE and islet-transplantation studies; mouse immune profiling.

PUBLICATIONS

18 TOTAL PAPERS	17 PEER-REVIEWED	1 IN REVISION	6 FIRST / CO-FIRST
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PUBLICATIONS · FIRST / CO-FIRST AUTHOR

- 2026 Woogil Song*, **Seong-Jun Kang***, Taehee Stella Kim*, Seyoung Jung, Sung Hee Kim, Su Chan Park, Sung Hae Chang, Eun Young Lee, Tae-Gyun Kim, Hyun Je Kim, Jeong Seok Lee. "Conserved PDE4D short-isoform bias defines pathogenic Th1.17 state across human inflammatory diseases." *Nature Immunology*. In revision
Co-first author. Developed UMI-aware, BAM-level isoform analysis across 251,055 memory CD4⁺ T cells, defined a pathogenic Th1.17 program, replicated it in 702 independent samples, and linked it to pharmacological inhibition.
- 2025 **Seong-Jun Kang***, Yong-Hee Kim*, Thuy Nguyen-Phuong*, Yijoon Kim, Jin-Mi Oh, Jae-chun Go, DaeSik Kim, Chung-Gyu Park, Hyunsu Lee, Hyun Je Kim. "Immune cell-enriched single-cell RNA sequencing unveils the interplay between infiltrated CD8⁺ T resident memory cells and choroid plexus epithelial cells in Alzheimer's disease." *Journal of Neuroimmunology*. Download PDF
Co-first author. Built the brain-to-single-cell path from whole-brain harvest through enzymatic digestion, Percoll enrichment and FACS to 10x; identified exhausted CD8⁺ tissue-resident memory T cells engaging choroid-plexus epithelium at the blood-CSF barrier.
- 2025 **Seong-Jun Kang***, Jeong-Ryeol Gong*, Seon-Pil Jin, Jin-Mi Oh, Hyunjin Jin, Yuji Lee, Yewon Moon, Dongjun Kim, Hyo Jeong Nam, Hyun Seung Choi, Sanha Hwang, Yun Jung Huh, Kyung Yeon Han, Jihwan Moon, Jongsuk Chung, Woong-Yang Park, Chung-Gyu Park, Hyun Je Kim, Jeong Eun Kim. "Deciphering Dysfunctional Regulatory T Cells in Atopic Dermatitis." *Allergy*. Download PDF
Co-first author. Single-cell and functional dissection of skin-homing regulatory T cells in atopic dermatitis, with OX40 up-regulated alongside impaired suppressive function.
- 2024 Jong-Min Kim*, **Seong-Jun Kang***, So-Hee Hong*, Hyunwoo Chung, Jun-Seop Shin, Byoung-Hoon Min, Hyun Je Kim, Jongwon Ha, Chung-Gyu Park. "Long-term control of diabetes by tofacitinib-based immunosuppressive regimen after allo-islet transplantation in diabetic rhesus monkeys that rejected previously transplanted porcine islets." *Xenotransplantation*. Download PDF
Co-first author. Long-term diabetes control with a tofacitinib-based regimen after allo-islet transplantation in rhesus macaques, with longitudinal immune monitoring.
- 2022 Youngkyoung Lim*, **Seong-Jun Kang***, Beom Keun Cho, Hyun Je Kim, ... Chong Hyun Won, Chung-Gyu Park. "Intra-tumoral heterogeneity and immune escape of melanoma arising from congenital melanocytic nevus revealed by spatial gene expression profiling." *JEADV*. Download PDF
Co-first author. Ran GeoMx DSP end to end on a nevus-arising melanoma: immunofluorescence staining, fluorescence-guided ROI selection, instrument operation, and the spatial-transcriptomic analysis across 16 depth-segmented regions.
- 2019 Jong-Min Kim*, Jun-Seop Shin*, Byoung-Hoon Min*, **Seong-Jun Kang***, Il-Hee Yoon, Hyunwoo Chung, Jiyeon Kim, Eung-Soo Hwang, Jongwon Ha, Chung-Gyu Park. "JAK3 inhibitor-based immunosuppression in allogeneic islet transplantation in cynomolgus monkeys." *Islets*. Download PDF
Co-first author, one of four authors contributing equally. JAK3-inhibitor-based immunosuppression in allogeneic islet transplantation in cynomolgus macaques.

CONTRIBUTING AUTHOR

- 2026 Jung Ho Lee*, Sung Ha Lim*, Hyungdon Kook, **Seong-Jun Kang**, Geon Lee, Brian Hyohyoung Lee, Hyunsung Nam, YongJun Kim, Christine Suh-Yun Joh, Soyoung Jeong, Dongryeol Shin, Hyun Je Kim, Jiyoung Ahn. "Mycosis Fungoides-like Atopic Dermatitis Represents a Th22-dominant Inflammatory Endotype." *Allergy*.
- 2024 Brian Hyohyoung Lee*, Yoon Ji Bang*, Sung Ha Lim*, **Seong-Jun Kang**, Sung Hee Kim, Seunghee Kim-Schulze, Chung-Gyu Park, Hyun Je Kim, Tae-Gyun Kim. "High-dimensional profiling of regulatory T cells in psoriasis reveals an impaired skin-trafficking property." *eBioMedicine*. Download PDF

2024	Youngkyoung Lim*, Beom Keun Cho*, Seong-Jun Kang , Soyoung Jeong, Hyun Je Kim, Jiyeon Baek, Ji Hwan Moon, Cheol Lee, Chan-Sik Park, Je-Ho Mun, Chong Hyun Won, Chung-Gyu Park. "Spatial transcriptomic analysis of tumour-immune cell interactions in melanoma arising from congenital melanocytic nevus." <i>JEADV</i> . Download PDF
2023	Sunyoung Jung*, Sunho Lee, Hyun Je Kim, Sueon Kim, Ji Hwan Moon, Hyunwoo Chung, Seong-Jun Kang , Chung-Gyu Park. "Mesenchymal stem cell-derived extracellular vesicles subvert Th17 cells by destabilizing RORγt through posttranslational modification." <i>Experimental & Molecular Medicine</i> .
2022	Sangho Lee, Han-Teo Lee, Young Ah Kim, Il-Hwan Lee, Seong-Jun Kang , Kyeongpyo Sim, Chung-Gyu Park, Kyungho Choi, Hong-Duk Youn. "The optimized core peptide derived from CABIN1 efficiently inhibits calcineurin-mediated T-cell activation." <i>Experimental & Molecular Medicine</i> .
2021	So Hee Hong, Hyun Je Kim, Seong-Jun Kang , Chung-Gyu Park. "Novel Immunomodulatory Approaches for Porcine Islet Xenotransplantation." <i>Current Diabetes Reports</i> . Download PDF
2020	Sunho Lee, Sueon Kim, Hyunwoo Chung, Ji Hwan Moon, Seong-Jun Kang , Chung-Gyu Park. "Mesenchymal stem cell-derived exosomes suppress proliferation of T cells by inducing cell cycle arrest through p27kip1/Cdk2 signaling." <i>Immunology Letters</i> .
2019	Hyun Je Kim, Ji Hwan Moon, Hyunwoo Chung, Jun-Seop Shin, Bongi Kim, Jong Min Kim, Jung-Sik Kim, Il-Hee Yoon, Byoung-Hoon Min, Seong-Jun Kang , Yong-Hee Kim, Kyuri Jo, Joungmin Choi, Heejoon Chae, Won Woo Lee, Sun Kim, Chung-Gyu Park. "Bioinformatic analysis of peripheral blood RNA-sequencing sensitively detects the cause of late graft loss following overt hyperglycemia in pig-to-non-human primate islet xenotransplantation." <i>Scientific Reports</i> .
2018	Byoung-Hoon Min, Jun-Seop Shin, Jong-Min Kim, Seong-Jun Kang , Hyun-Je Kim, Il-Hee Yoon, Su-Kyoung Park, Ji-won Choi, Min-Suk Lee, Chung-Gyu Park. "Delayed revascularization of islets after transplantation by IL-6 blockade in pig-to-non-human primate islet xenotransplantation model." <i>Xenotransplantation</i> .
2018	Jun-Seop Shin, Jong-Min Kim, Byoung-Hoon Min, Il Hee Yoon, Hyun Je Kim, Jung-Sik Kim, Yong-Hee Kim, Seong-Jun Kang , Jiyeon Kim, Hee-Jung Kang, Dong-Gyun Lim, Eung-Soo Hwang, Jongwon Ha, Sang-Joon Kim, Wan Beom Park, Chung-Gyu Park. "Pre-clinical results in pig-to-non-human primate islet xenotransplantation using anti-CD40 antibody (2C10R4)-based immunosuppression." <i>Xenotransplantation</i> .
2018	Jong-Min Kim, Jun-Seop Shin, Byoung-Hoon Min, Il Hee Yoon, Seong-Jun Kang , Won-Young Jeong, Sang-Joon Kim, Chung-Gyu Park. "Pre-Clinical Results of Islet Allo-Transplantation Using JAK Inhibitor as Replacement for Tacrolimus in Cynomolgus Monkeys." <i>Transplantation</i> .
2018	Jung-Sik Kim, Hyunwoo Chung, Nari Byun, Seong-Jun Kang , Sunho Lee, Jun-Seop Shin, Chung-Gyu Park. "Construction of EMSC-islet co-localizing composites for xenogeneic porcine islet transplantation." <i>Biochemical and Biophysical Research Communications</i> .

MANUSCRIPTS IN PREPARATION

- 2026 Kangsan Roh*, Sabyasachi Das*, **Seong-Jun Kang**, Olga Barreiro, Rodrigo Javier Gonzalez, Timothy P. Padera, Shraddha Shirguppe, Ahlim Lee, Youngju Song, Pazhanichamy Kalailingam, Claire E. Fong, Aarushi Gandhi, Meehyun Kim, Diana Tambala, Adam Vernon Crain, Casey A. Maguire, Ulrich H. Von Andrian, Christian L. Lino Cardenas, Sekeun Kim, Benjamin P. Kleinstiver, Mark E. Lindsay, Patricia L. Musolino, Rajeev Malhotra. "In-vivo base editing of ACTA2 R179H restores lymphatic function and immune cell homeostasis in a murine model of Multisystemic Smooth Muscle Dysfunction Syndrome (MSMDS)." [in preparation](#)
Co-author (immunology). Ran the Olink plasma proteomics across 12 patients and 6 controls and the flow-cytometry immunophenotyping of lymph-node leukocytes.

■ **SELECTED GRANT WRITING & PROGRAM FUNDING**

INDUSTRY · PB IMMUNE THERAPEUTICS (MANAGER – ASSOCIATE DIRECTOR)

FUNDING AGENCY / SPONSOR	PROJECT	PERIOD	BUDGET (USD)	ROLE / CONTRIBUTION
Korea Drug Development Fund (KDDF)	Anti-CD40 antibody lead selection and optimization (multiple sclerosis)	2021–2023	~\$571,000	Lead proposal writer (>90% of proposal & execution)
Korea Drug Development Fund (KDDF)	Anti-CD40 lead compound (PB101): efficacy and pharmacodynamic biomarkers (pemphigus vulgaris)	2022–2024	~\$321,000	Lead proposal writer (>90% of proposal & execution)
Ministry of SMEs and Startups (MSS)	Scale-up Funding for Tech Commercialization	2022–2023	~\$222,000	Lead proposal writer; managed >90% of execution
Ministry of SMEs and Startups (MSS)	Early-stage startup R&D: anti-CD40 biologic (PB-40N) to prevent anti-drug antibodies	2021–2022	~\$74,000	Lead proposal writer & core researcher
NAVER Corporation	Engineered regulatory T cells for immune-mediated diseases (PB20X_Treg)	2023–2026	~\$128,000	Proposal & study design (~50%)

DOCTORAL TRAINING · SEOUL NATIONAL UNIVERSITY

FUNDING AGENCY / SPONSOR	PROJECT	PERIOD	BUDGET (USD)	ROLE / CONTRIBUTION
Ministry of Science and ICT (MSIT)	Multi-layered immune-disease single-cell map for molecular target discovery (SCAID, A9-22-1-05)	2022–2028 (3+3)	~\$1.6M	Participated in proposal writing (>30%)
Ministry of Education (MOE) & Ministry of Science and ICT (MSIT)	Laboratory-Specialized Tech-Transfer & R&BD Grant	2020–2021	~\$40,000	Grant writing, experimental design & technical validation

All grants were held by the institutional Principal Investigator (Ph.D. advisor at SNU; company CEO/CSO at PB Immune Therapeutics); the listed roles reflect the applicant's contribution to proposal writing and project execution, **not as PI**. Budgets converted at ≈₩1,400 / USD.

■ **RESEARCH INTERESTS**

- **Reading immune cell state from incomplete single-cell data.** Cell-state calling under ambiguity, isoform and transcript resolution from short reads, and embedding representations of cells, with calibrated confidence.
- **Multi-modal integration of molecular identity.** Transcriptome, chromatin, spatial, and proteome into cell-state-resolved models.
- **Neuro-immune interactions in neurodegeneration.** T cells, microglia, and the blood-CSF barrier in Alzheimer's and related disease; ligand-receptor cross-talk and clonal/repertoire structure.
- **Building and checking computational analysis paths.** Custom callers, reproducible end-to-end analysis code, and comparison against domain ground truth.
- **Atlas-scale single-cell analysis across diseases and species.** Large multi-donor atlases (SCAID), cross-disease meta-analysis, and non-human-primate-to-human translation.

■ **AWARDS, PATENTS & SCHOLARSHIPS**

PATENTS

2024 **Antagonistic Anti-CD40 Humanized Antibody.** Republic of Korea, Appl. No. 10-2024-0085972; registered 2024-07-01. Co-inventor.

AWARDS

2023 **Grand Prize**, SBA Smart Workathon (Work Process Innovation)

SCHOLARSHIPS

■ SELECTED PRESENTATIONS

REFERENCES